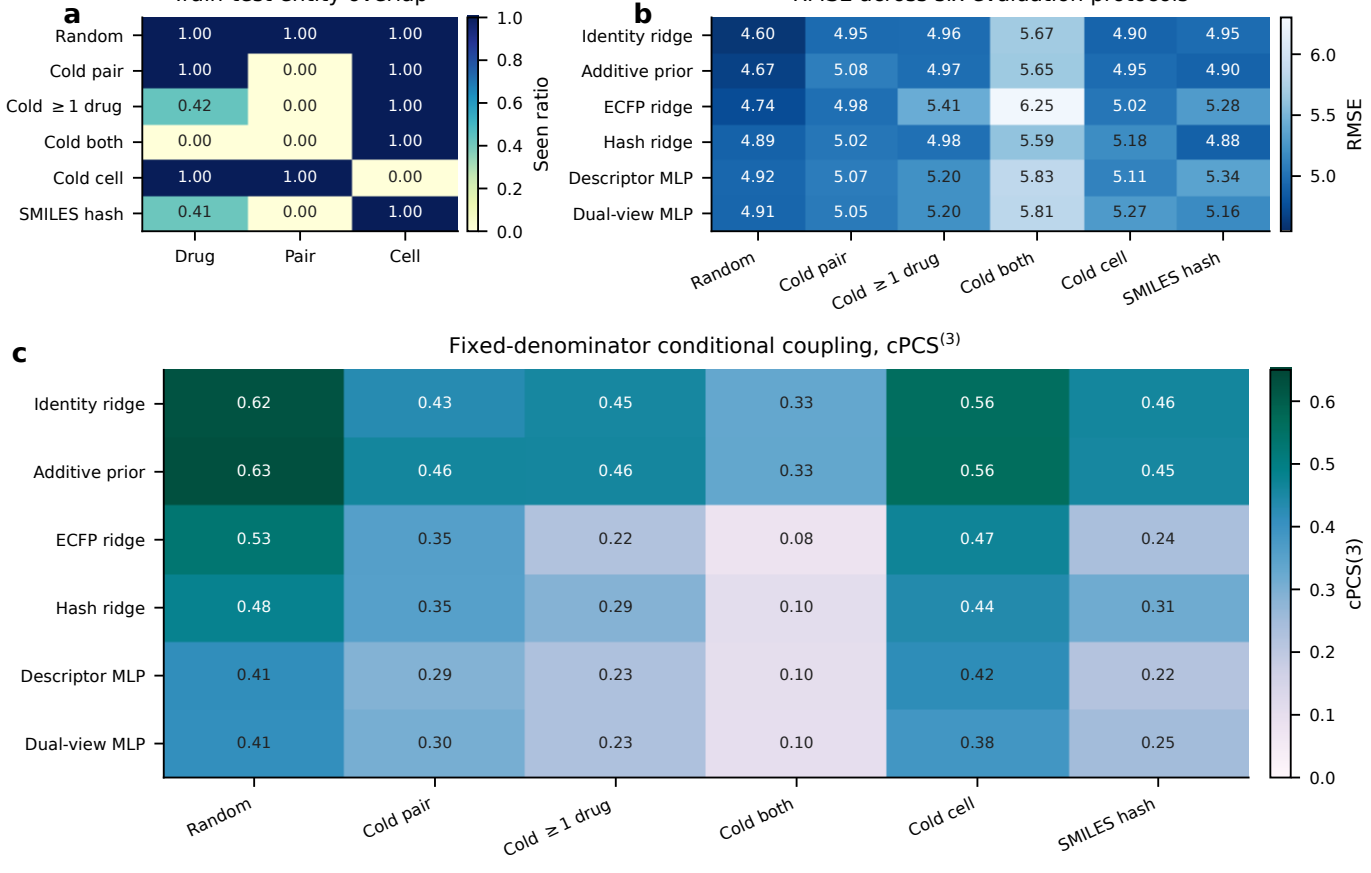
a
Train-test entity overlap
Random
Cold pair
Cold ≥ 1 drug
Cold both
Cold cell
SMILES hash
1.00 1.00 1.00
1.00 0.00 1.00
0.42 0.00 1.00
0.00 0.00 1.00
1.00 1.00 0.00
0.41 0.00 1.00
Drug
Pair
Cell
Seen ratio
1.0
0.8
0.6
0.4
0.2
0.0
b
RMSE across six evaluation protocols
Identity ridge
Additive prior
ECFP ridge
Hash ridge
Descriptor MLP
Dual-view MLP
4.60 4.95 4.96 5.67 4.90 4.95
4.67 5.08 4.97 5.65 4.95 4.90
4.74 4.98 5.41 6.25 5.02 5.28
4.89 5.02 4.98 5.59 5.18 4.88
4.92 5.07 5.20 5.83 5.11 5.34
4.91 5.05 5.20 5.81 5.27 5.16
Random
Cold pair
Cold ≥ 1 drug
Cold both
Cold cell
SMILES hash
RMSE
6.0
5.5
5.0
c
Fixed-denominator conditional coupling, cPCS(3)
Identity ridge
Additive prior
ECFP ridge
Hash ridge
Descriptor MLP
Dual-view MLP
0.62 0.43 0.45 0.33 0.56 0.46
0.63 0.46 0.46 0.33 0.56 0.45
0.53 0.35 0.22 0.08 0.47 0.24
0.48 0.35 0.29 0.10 0.44 0.31
0.41 0.29 0.23 0.10 0.42 0.22
0.41 0.30 0.23 0.10 0.38 0.25
Random
Cold pair
Cold ≥ 1 drug
Cold both
Cold cell
SMILES hash
cPCS(3)
0.6
0.5
0.4
0.3
0.2
0.1
0.0